

Multiple Choice

1. **Answer:** B

Options: A) The resonance pulls Io in different directions and generates heat.; B) It makes the orbit of Io slightly elliptical.; C) It creates a gap with no asteroids between the orbits.; D) It prevents formation of the ring material into other moons.

Note: Correct option: B - It makes the orbit of Io slightly elliptical.

2. **Answer:** D

Options: A) because the Moon has a nearly circular orbit around the Earth; B) because the Moon does not rotate; C) because the other face points toward us only at new moon when we can't see the Moon; D) because the Moon's rotational and orbital periods are equal

Note: Correct option: D - because the Moon's rotational and orbital periods are equal

3. **Answer:** B

Options: A) Spontaneous emission from radioactive atoms.; B) The minimization of gravitational potential energy.; C) Thermally induced collisions.; D) Plate tectonics.

Note: Correct option: B - The minimization of gravitational potential energy.

4. **Answer:** D

Options: A) fast rotation only; B) a rocky mantle only; C) a molten metallic core only; D) both a molten metallic core and reasonably fast rotation

Note: Correct option: D - both a molten metallic core and reasonably fast rotation

5. **Answer:** D

Options: A) hydrogen compounds (H₂O CH₄ NH₃) light gases (H He) metals rocks; B) hydrogen compounds (H₂O CH₄ NH₃) light gases (H He) rocks metals; C) light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) metals rocks; D) light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) rocks metals

Note: Correct option: D - light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) rocks metals

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Sagittarius A*'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Radioactive dating of lunar samples shows that they are older.'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'third quarter.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Jupiter. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.5×10^{17} meters. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key